RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

Laboratory Record Note Book

Name :	Bharath S
Year / Branch / Section :	II Year / CSD / A
University Register No. :	2116251701009
College Roll No. :	251701009
Semester :	III
Academic Year :	2026 - 2027

Started on	Monday, 3 August 2026, 4:01 PM
State	Finished
Completed on	Monday, 3 August 2026, 4:13 PM
Time taken	11 mins 20 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

Reset answer

```
#include<stdio.h>

int c = 0;

void func(int n){
    int i=1;
    c++;
    int s=1;
    c++;
    while(s<=n){
        c++;
        i++;
        c++;
        s += i;
        c++;
    }

    c++;
}

int main(){
    int n;
    scanf("%d", &n);
    func(n);
    printf("%d", c);
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Monday, 3 August 2026, 4:15 PM
State	Finished
Completed on	Monday, 3 August 2026, 4:29 PM
Time taken	14 mins 54 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

Reset answer

```
#include<stdio.h>

int c = 0;

void func(int n)
{
    c++;
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       c++;
       for(int j=1; j<=n; j++)
       {
          c++;
          c++;
          c++;
          //printf("*");
          //printf("*");
          break;
          c++;
       }
       c++;
     }
     c++;
   }
 }
```

```
int main(){
    int n;
    scanf("%d",&n);
    func(n);
    printf("%d", c);
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Sunday, 9 August 2026, 1:39 PM
State	Finished
Completed on	Sunday, 9 August 2026, 1:48 PM
Time taken	8 mins 56 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Reset answer

```c
#include <stdio.h>

int main(){
    int num,i,c = 0;
    scanf("%d", &num);
    c++;
    for (i = 1; i <= num;i++)
    {
        c++;
        if (num % i== 0)
        {
            c++;
        }
        c++;
    }
    printf("%d", c);
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Wednesday, 26 August 2026, 11:11 AM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:16 AM
Time taken	4 mins 34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Reset answer

```
#include <stdio.h>
int main()
{
    int c= 0;
int n;
c++;
scanf("%d", &n);
    for(int i=n/2; i<n; i++){
    c++;
        for(int j=1; j<n; j = 2 * j){
        c++;
            for(int k=1; k<n; k = k * 2){
            c++;
                c++;
            }
            c++;
        }
        c++;
    }
    c++;

printf("%d", c);
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Wednesday, 26 August 2026, 11:18 AM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:21 AM
Time taken	3 mins 20 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

Reset answer

```
#include <stdio.h>
int main()
{
   int c = 0;
   int n;
   scanf("%d",&n);
   c++;
   int rev = 0;
   c++;
   int remainder;
   c++;
   while (n != 0)
    {
        c++;
        remainder = n % 10;
        c++;
        rev = rev * 10 + remainder;
        c++;
        n/= 10;
        c++;
    }
printf("%d",c);
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.